

MATHEMATICS

JEE (MAIN+ADVANCED)

NURTURE COURSE

$$S_n = n \left(\frac{a_1 + a_n}{2} \right)$$

Study Material

Sequence & Series

(English Medium)

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SEQUENCE & SERIES

1. DEFINITION :

Sequence :

A succession of terms $a_1, a_2, a_3, a_4, \dots$ formed according to some rule or law.

Examples are : 1, 4, 9, 16, 25

$-1, 1, -1, 1, \dots$

$$\frac{x}{1!}, \frac{x^2}{2!}, \frac{x^3}{3!}, \frac{x^4}{4!}, \dots$$

A finite sequence has a finite (i.e. limited) number of terms, as in the first example above. An infinite sequence has an unlimited number of terms, i.e. there is no last term, as in the second and third examples.

Series :

The indicated sum of the terms of a sequence. In the case of a finite sequence $a_1, a_2, a_3, \dots, a_n$

the corresponding series is $a_1 + a_2 + a_3 + \dots + a_n = \sum_{k=1}^n a_k$. This series has a finite or limited number of terms and is called a finite series.

2. ARITHMETIC PROGRESSION (A.P.) :

A.P. is a sequence whose terms differ by a fixed number. This fixed number is called the common difference. If a is the first term & d the common difference, then A.P. can be written as

$$a, a + d, a + 2d, \dots, a + (n-1)d, \dots$$

(a) n^{th} term of AP $T_n = a + (n-1)d$, where $d = t_n - t_{n-1}$

(b) The sum of the first n terms : $S_n = \frac{n}{2}[a + \ell] = \frac{n}{2}[2a + (n-1)d]$

where ℓ is n^{th} term.

Note :

- (i) n^{th} term of an A.P. is of the form $An + B$ i.e. a linear expression in ' n ', in such a case the coefficient of n is the common difference of the A.P. i.e. A .
- (ii) Sum of first ' n ' terms of an A.P. is of the form $An^2 + Bn$ i.e. a quadratic expression in ' n ', in such case the common difference is twice the coefficient of n^2 . i.e. $2A$
- (iii) Also n^{th} term $T_n = S_n - S_{n-1}$

Illustration 1 : If $(x + 1)$, $3x$ and $(4x + 2)$ are first three terms of an A.P. then its 5th term is -

- (A) 14 (B) 19 (C) 24 (D) 28

Solution : $(x + 1)$, $3x$, $(4x + 2)$ are in AP

$$\Rightarrow 3x - (x + 1) = (4x + 2) - 3x \quad \Rightarrow x = 3$$

$$\therefore a = 4, d = 9 - 4 = 5 \quad \Rightarrow T_5 = 4 + (4)5 = 24 \quad \text{Ans. (C)}$$

Illustration 2 : The sum of first four terms of an A.P. is 56 and the sum of its last four terms is 112. If its first term is 11 then find the number of terms in the A.P.

Solution : $a + a + d + a + 2d + a + 3d = 56$

$$4a + 6d = 56$$

$$44 + 6d = 56 \quad (\text{as } a = 11)$$

$$6d = 12 \quad \text{hence } d = 2$$

Let total number of terms = n

Now sum of last four terms.

$$a + (n - 1)d + a + (n - 2)d + a + (n - 3)d + a + (n - 4)d = 112$$

$$\Rightarrow 4a + (4n - 10)d = 112 \quad \Rightarrow 44 + (4n - 10)2 = 112$$

$$\Rightarrow 4n - 10 = 34$$

$$\Rightarrow n = 11$$

Ans.

Illustration 3 : The sum of first n terms of two A.P.s. are in ratio $\frac{7n+1}{4n+27}$. Find the ratio of their 11th terms.

Solution : Let a_1 and a_2 be the first terms and d_1 and d_2 be the common differences of two A.P.s respectively then

$$\frac{\frac{n}{2}[2a_1 + (n-1)d_1]}{\frac{n}{2}[2a_2 + (n-1)d_2]} = \frac{7n+1}{4n+27} \Rightarrow \frac{a_1 + \left(\frac{n-1}{2}\right)d_1}{a_2 + \left(\frac{n-1}{2}\right)d_2} = \frac{7n+1}{4n+27}$$

For ratio of 11th terms

$$\frac{n-1}{2} = 10 \quad \Rightarrow n = 21$$

$$\text{so ratio of 11th terms is } \frac{7(21)+1}{4(21)+27} = \frac{148}{111} = \frac{4}{3}$$

Ans.

Do yourself - 1 :

1. Write down the sequence whose n^{th} terms is : (a) $\frac{2^n}{n}$ (b) $\frac{3+(-1)^n}{3^n}$
2. For an A.P, show that $t_m + t_{2n+m} = 2t_{m+n}$
3. If the sum of first p terms of an A.P. is q and the sum of its first q terms is p , then find the sum of its first $(p+q)$ term.

Find the

4. sum 49, 44, 39.... to 17 terms.
5. sum $\frac{3}{4}, \frac{2}{3}, \frac{7}{12}, \dots$ to 19 terms.
6. sum 1.3, -3.1, -7.5,.... to 10 terms
7. sum $a - 3b, 2a - 5b, 3a - 7b, \dots$ to 40 terms.
8. In an AP of which 'a' is the 1st term, if the sum of the 1st p terms is equal to zero, show that the sum of the next q terms is $-\left(\frac{aq(p+q)}{p-1}\right)$

3. PROPERTIES OF A.P. :

- (a) If each term of an A.P. is increased, decreased, multiplied or divided by the some nonzero number, then the resulting sequence is also an A.P.
- (b) Three numbers in A.P. : $a-d, a, a+d$
Four numbers in A.P. : $a-3d, a-d, a+d, a+3d$
Five numbers in A.P. : $a-2d, a-d, a, a+d, a+2d$
Six numbers in A.P. : $a-5d, a-3d, a-d, a+d, a+3d, a+5d$ etc.
- (c) The common difference can be zero, positive or negative.
- (d) k^{th} term from the last $= (n-k+1)^{\text{th}}$ term from the beginning (If total number of terms $= n$).
- (e) The sum of the two terms of an AP equidistant from the beginning & end is constant and equal to the sum of first & last terms. $\Rightarrow T_k + T_{n-k+1} = \text{constant} = a + \ell$.
- (f) Any term of an AP (except the first) is equal to half the sum of terms which are equidistant from it. $a_n = (1/2)(a_{n-k} + a_{n+k}), k < n$

For $k=1$, $a_n = (1/2)(a_{n-1} + a_{n+1})$; For $k=2$, $a_n = (1/2)(a_{n-2} + a_{n+2})$ and so on.

- (g) If a, b, c are in AP, then $2b = a + c$.

Illustration 4 : Four numbers are in A.P. If their sum is 20 and the sum of their squares is 120, then the middle terms are -

- (A) 2, 4 (B) 4, 6 (C) 6, 8 (D) 8, 10

Solution : Let the numbers are $a - 3d, a - d, a + d, a + 3d$

$$\text{given, } a - 3d + a - d + a + d + a + 3d = 20 \quad \Rightarrow 4a = 20 \Rightarrow a = 5$$

$$\text{and } (a - 3d)^2 + (a - d)^2 + (a + d)^2 + (a + 3d)^2 = 120 \quad \Rightarrow 4a^2 + 20d^2 = 120$$

$$\Rightarrow 4 \times 5^2 + 20d^2 = 120 \quad \Rightarrow d^2 = 1 \Rightarrow d = \pm 1$$

Hence numbers are 2, 4, 6, 8 or 8, 6, 4, 2 **Ans. (B)**

Illustration 5 : If $a_1, a_2, a_3, \dots, a_n$ are in A.P. where $a_i > 0$ for all i , show that :

$$\frac{1}{\sqrt{a_1} + \sqrt{a_2}} + \frac{1}{\sqrt{a_2} + \sqrt{a_3}} + \dots + \frac{1}{\sqrt{a_{n-1}} + \sqrt{a_n}} = \frac{(n-1)}{\sqrt{a_1} + \sqrt{a_n}}$$

Solution :

$$\text{L.H.S.} = \frac{1}{\sqrt{a_1} + \sqrt{a_2}} + \frac{1}{\sqrt{a_2} + \sqrt{a_3}} + \dots + \frac{1}{\sqrt{a_{n-1}} + \sqrt{a_n}}$$

$$= \frac{1}{\sqrt{a_2} + \sqrt{a_1}} + \frac{1}{\sqrt{a_3} + \sqrt{a_2}} + \dots + \frac{1}{\sqrt{a_n} + \sqrt{a_{n-1}}}$$

$$= \frac{\sqrt{a_2} - \sqrt{a_1}}{(a_2 - a_1)} + \frac{\sqrt{a_3} - \sqrt{a_2}}{(a_3 - a_2)} + \dots + \frac{\sqrt{a_n} - \sqrt{a_{n-1}}}{a_n - a_{n-1}}$$

Let 'd' is the common difference of this A.P.

$$\text{then } a_2 - a_1 = a_3 - a_2 = \dots = a_n - a_{n-1} = d$$

Now L.H.S.

$$= \frac{1}{d} \{ \sqrt{a_2} - \sqrt{a_1} + \sqrt{a_3} - \sqrt{a_2} + \dots + \sqrt{a_{n-1}} - \sqrt{a_{n-2}} + \sqrt{a_n} - \sqrt{a_{n-1}} \} = \frac{1}{d} \{ \sqrt{a_n} - \sqrt{a_1} \}$$

$$= \frac{a_n - a_1}{d(\sqrt{a_n} + \sqrt{a_1})} = \frac{a_1 + (n-1)d - a_1}{d(\sqrt{a_n} + \sqrt{a_1})} = \frac{1}{d} \frac{(n-1)d}{(\sqrt{a_n} + \sqrt{a_1})} = \frac{n-1}{\sqrt{a_n} + \sqrt{a_1}} = \text{R.H.S.}$$

Illustration 6 : The series of natural numbers is divided into groups (1), (2, 3, 4), (5, 6, 7, 8, 9) and so on. Show that the sum of the numbers in n^{th} group is $n^3 + (n - 1)^3$

Solution : The groups are (1), (2, 3, 4), (5, 6, 7, 8, 9)

The number of terms in the groups are 1, 3, 5.....

$\therefore$ The number of terms in the n^{th} group = $(2n - 1)$

the last term of the n^{th} group is n^2

If we count from last term common difference should be -1

$$\text{So the sum of numbers in the } n^{\text{th}} \text{ group} = \left(\frac{2n-1}{2} \right) \{ 2n^2 + (2n-2)(-1) \}$$

$$= (2n-1)(n^2 - n + 1) = 2n^3 - 3n^2 + 3n - 1 = n^3 + (n-1)^3$$

Do yourself - 2 :

- Find the sum of first 24 terms of the A.P. $a_1, a_2, a_3, \dots$, if it is known that $a_1 + a_5 + a_{10} + a_{15} + a_{20} + a_{24} = 225$.
- Find the number of terms common to the two A.P.'s 3, 7, 11, 407 and 2, 9, 16, 709
- In an A.P. the first term is 2, the last term is 29, the sum is 155; find the common difference.
- The sum of 15 terms of an A.P. is 600, and the common difference is 5; find the first term.
- The third term of an A.P. is 18, and the seventh term is 30; find the sum of 17 terms.
- The sum of three numbers in A.P. is 27, and their product is 504; find them.
- The sum of three numbers in A.P. is 12, and the sum of their cubes is 408; find them.
- Find the sum of 15 terms of the series whose n^{th} term is $4n + 1$.
- In an AP if the sum of 7 terms is 49, and the sum of 17 terms is 289, find the sum of n terms.
- The sum of four integers in A.P. is 24, and their product is 945; find them.
- If the sum of n terms of an A.P. is $2n + 3n^2$, find the r^{th} term.
- If the sum of m terms of an A.P. is to the sum of n terms as m^2 to n^2 , show that the m^{th} term is to the n^{th} term as $2m - 1$ is to $2n - 1$.
- The sum of n terms of two arithmetic series are in the ratio of $(7n + 1) : (4n + 27)$. Find the ratio of their n^{th} term.
- If the sum of n terms of an AP is $Pn + Qn^2$, where P, Q are constants, then its common difference is

4. GEOMETRIC PROGRESSION (G.P.) :

G.P. is a sequence of non zero numbers each of the succeeding term is equal to the preceeding term multiplied by a constant. Thus in a GP the ratio of successive terms is constant. This constant factor is called the COMMON RATIO of the sequence & is obtained by dividing any term by the immediately previous term. Therefore $a, ar, ar^2, ar^3, ar^4, \dots$ is a GP with 'a' as the first term & 'r' as common ratio.

(a) n^{th} term ; $T_n = a r^{n-1}$

(b) Sum of the first n terms; $S_n = \frac{a(r^n - 1)}{r - 1}$, if $r \neq 1$

(c) Sum of infinite G.P. , $S_\infty = \frac{a}{1 - r}$; $0 < |r| < 1$

5. PROPERTIES OF GP :

- If each term of a G.P. be multiplied or divided by the some non-zero quantity, then the resulting sequence is also a G.P.
- Three consecutive terms of a GP : $a/r, a, ar$;
Four consecutive terms of a GP : $a/r^3, a/r, ar, ar^3$ & so on.
- If a, b, c are in G.P. then $b^2 = ac$.
- If in a G.P, the product of two terms which are equidistant from the first and the last term, is constant and is equal to the product of first and last term. $\Rightarrow T_k \cdot T_{n-k+1} = \text{constant} = a \cdot \ell$
- If each term of a G.P. be raised to the same power, then resulting sequence is also a G.P.
- In a G.P., $T_r^2 = T_{r-k} \cdot T_{r+k}$, $k < r$, $r \neq 1$
- If the terms of a given G.P. are chosen at regular intervals, then the new sequence is also a G.P.
- If $a_1, a_2, a_3, \dots, a_n$ is a G.P. of positive terms, then $\log a_1, \log a_2, \dots, \log a_n$ is an A.P. and vice-versa.
- If $a_1, a_2, a_3, \dots$ and $b_1, b_2, b_3, \dots$ are two G.P.'s then $a_1 b_1, a_2 b_2, a_3 b_3, \dots$ & $\frac{a_1}{b_1}, \frac{a_2}{b_2}, \frac{a_3}{b_3}, \dots$ is also in G.P.

Illustration 7 : If a, b, c, d and p are distinct real numbers such that

$$(a^2 + b^2 + c^2)p^2 - 2p(ab + bc + cd) + (b^2 + c^2 + d^2) \leq 0 \text{ then } a, b, c, d \text{ are in}$$

- (A) A.P. (B) G.P. (C) H.P. (D) None of these

Solution : Here, the given condition $(a^2 + b^2 + c^2)p^2 - 2p(ab + bc + cd) + b^2 + c^2 + d^2 \leq 0$

$$\Rightarrow (ap - b)^2 + (bp - c)^2 + (cp - d)^2 \leq 0$$

$\therefore$ a square can not be negative

$$\therefore ap - b = 0, bp - c = 0, cp - d = 0 \Rightarrow p = \frac{b}{a} = \frac{c}{b} = \frac{d}{c} \Rightarrow a, b, c, d \text{ are in G.P.}$$

Ans. (B)

Illustration 8 : If positive real numbers a, b, c are in G.P., then the equations $ax^2 + 2bx + c = 0$ and

$dx^2 + 2ex + f = 0$ have a common root if $\frac{d}{a}, \frac{e}{b}, \frac{f}{c}$ are in -

- (A) A.P. (B) G.P. (C) H.P. (D) None of these

Solution : a, b, c are in G.P. $\Rightarrow b^2 = ac$

Now the equation $ax^2 + 2bx + c = 0$ can be rewritten as $ax^2 + 2\sqrt{ac}x + c = 0$

$$\Rightarrow (\sqrt{a}x + \sqrt{c})^2 = 0 \Rightarrow x = -\sqrt{\frac{c}{a}}, -\sqrt{\frac{c}{a}}$$

If the two given equations have a common root, then this root must be $-\sqrt{\frac{c}{a}}$.

$$\text{Thus } d\frac{c}{a} - 2e\sqrt{\frac{c}{a}} + f = 0 \Rightarrow \frac{d}{a} + \frac{f}{c} = \frac{2e}{c}\sqrt{\frac{c}{a}} = \frac{2e}{\sqrt{ac}} = \frac{2e}{b} \Rightarrow \frac{d}{a}, \frac{e}{b}, \frac{f}{c} \text{ are in A.P.}$$

Ans. (A)

Illustration 9 : A number consists of three digits which are in G.P. the sum of the right hand and left hand digits exceeds twice the middle digit by 1 and the sum of the left hand and middle digits is two third of the sum of the middle and right hand digits. Find the numbers.

Solution : Let the three digits be a, ar and ar^2 then number is

$$100a + 10ar + ar^2 \quad \dots(i)$$

$$\text{Given, } a + ar^2 = 2ar + 1$$

$$\text{or } a(r^2 - 2r + 1) = 1$$

$$\text{or } a(r - 1)^2 = 1 \quad \dots(ii)$$

$$\text{Also given } a + ar = \frac{2}{3}(ar + ar^2)$$

$$\Rightarrow 3 + 3r = 2r + 2r^2 \Rightarrow 2r^2 - r - 3 = 0 \Rightarrow (r + 1)(2r - 3) = 0$$

$$\therefore r = -1, 3/2$$

$$\text{for } r = -1, \quad a = \frac{1}{(r-1)^2} = \frac{1}{4} \notin \mathbb{I} \quad \therefore r \neq -1$$

$$\text{for } r = 3/2, \quad a = \frac{1}{\left(\frac{3}{2}-1\right)^2} = 4 \quad \{\text{from (ii)}\}$$

$$\text{From (i), number is } 400 + 10 \cdot 4 \cdot \frac{3}{2} + 4 \cdot \frac{9}{4} = 469$$

Ans.

Illustration 10 : Find the value of $0.32\overline{58}$

Solution : Let $R = 0.32\overline{58} \Rightarrow R = 0.32585858\dots$ (i)

Here number of figures which are not recurring is 2 and number of figures which are recurring is also 2.

$$\text{then } 100 R = 32.585858\dots \quad \dots\dots(ii)$$

$$\text{and } 10000 R = 3258.5858\dots \quad \dots\dots(iii)$$

Subtracting (ii) from (iii), we get

$$9900 R = 3226 \Rightarrow R = \frac{1613}{4950}$$

Aliter Method : $R = .32 + .0058 + .000058 + .00000058 + \dots\dots\dots$

$$\begin{aligned} &= .32 + \frac{58}{10^4} \left(1 + \frac{1}{10^2} + \frac{1}{10^4} + \dots\dots\dots\infty \right) = .32 + \frac{58}{10^4} \left(\frac{1}{1 - \frac{1}{100}} \right) \\ &= \frac{32}{100} + \frac{58}{9900} = \frac{3168 + 58}{9900} = \frac{3226}{9900} = \frac{1613}{4950} \end{aligned}$$

Do yourself - 3 :

- Find a three digit number whose consecutive digits form a G.P. If we subtract 792 from this number, we get a number consisting of the same digits written in the reverse order.
- If the third term of G.P. is 4, then find the product of first five terms.
- If a, b, c are respectively the p^{th} , q^{th} and r^{th} terms of the given G.P., then show that $(q - r) \log a + (r - p) \log b + (p - q) \log c = 0$, where a, b, c > 0.
- Find three numbers in G.P., whose sum is 52 and the sum of whose products in pairs is 624.
- The rational number which equals the number $2.\overline{357}$ with recurring decimal is -

- (A) $\frac{2357}{999}$ (B) $\frac{2379}{997}$ (C) $\frac{785}{333}$ (D) $\frac{2355}{1001}$

Find the

- sum $\frac{3}{4}, 1\frac{1}{2}, 3, \dots$ to 8 terms.
- sum $1, 5, 25, \dots$ to p terms.
- sum $1, \sqrt{3}, 3, \dots$ to 12 terms.
- sum $\frac{8}{5}, -1, \frac{5}{8}, \dots$ upto infinity
- sum $.45, .015, .0005, \dots$ upto infinity
- sum $3, \sqrt{3}, 1, \dots$ upto infinity
- The sum of the first 6 terms of a G.P. is 9 times the sum of the first 3 terms; find the common ratio.

13. The fifth term of a G.P. is 81, and the second term is 24; find the series.
14. The sum of three numbers in G.P. is 38, and their product is 1728; find them.
15. The continued product of three numbers in G.P. is 216, and the sum of the product of them in pairs is 156; find the numbers.
16. The sum of an infinite number of terms of a G.P. is 4, and the sum of their cubes is 192; find the series.
17. The first two terms of an infinite G.P. are together equal to 5, and every term is 3 times the sum of all the terms that follow it; find the series.
18. Sum of following series : $x + a, x^2 + 2a, x^3 + 3a, \dots$ to n terms.
19. In a GP the ratio of the sum of the first eleven terms to the sum of the last eleven terms is $1/8$ and the ratio of the sum of all the terms without the first nine to the sum of all the terms without the last nine is 2. Find the number of terms in the GP.

6. HARMONIC PROGRESSION (H.P.) :

A sequence is said to be in H.P. if the reciprocal of its terms are in AP.

If the sequence $a_1, a_2, a_3, \dots, a_n$ is an HP then $1/a_1, 1/a_2, \dots, 1/a_n$ is an AP. Here we do not have the formula for the sum of the n terms of an HP. The general form of a harmonic progression is

$$\frac{1}{a}, \frac{1}{a+d}, \frac{1}{a+2d}, \dots, \frac{1}{a+(n-1)d}$$

Note : No term of any H.P. can be zero.

$$(i) \quad \text{If } a, b, c \text{ are in HP, then } b = \frac{2ac}{a+c} \text{ or } \frac{a}{c} = \frac{a-b}{b-c}$$

Illustration 11 : The sum of three numbers are in H.P. is 37 and the sum of their reciprocals is $1/4$. Find the numbers.

Solution : Three numbers are in H.P. can be taken as

$$\frac{1}{a-d}, \frac{1}{a}, \frac{1}{a+d}$$

$$\text{then } \frac{1}{a-d} + \frac{1}{a} + \frac{1}{a+d} = 37 \quad \dots\dots(i)$$

$$\text{and } a-d + a + a+d = \frac{1}{4} \quad \Rightarrow \quad a = \frac{1}{12}$$

$$\text{from (i), } \frac{12}{1-12d} + 12 + \frac{12}{1+12d} = 37 \quad \Rightarrow \quad \frac{12}{1-12d} + \frac{12}{1+12d} = 25$$

$$\Rightarrow \frac{24}{1-144d^2} = 25$$

$$\Rightarrow 1-144d^2 = \frac{24}{25}$$

$$\Rightarrow d^2 = \frac{1}{25 \times 144}$$

$$\therefore d = \pm \frac{1}{60}$$

$$\therefore a-d, a, a+d \text{ are } \frac{1}{15}, \frac{1}{12}, \frac{1}{10} \text{ or } \frac{1}{10}, \frac{1}{12}, \frac{1}{15}$$

Hence, three numbers in H.P. are 15, 12, 10 or 10, 12, 15

Ans.

Illustration 12 : Suppose a is a fixed real number such that $\frac{a-x}{px} = \frac{a-y}{qy} = \frac{a-z}{rz}$

If p, q, r are in A.P., then prove that x, y, z are in H.P.

Solution :

$\therefore p, q, r$ are in A.P.

$$\therefore q-p = r-q \quad \dots\dots (i)$$

$$\Rightarrow p-q = q-r = k \text{ (let)}$$

$$\text{given } \frac{a-x}{px} = \frac{a-y}{qy} = \frac{a-z}{rz} \Rightarrow \frac{\frac{a}{x}-1}{p} = \frac{\frac{a}{y}-1}{q} = \frac{\frac{a}{z}-1}{r}$$

$$\Rightarrow \frac{\left(\frac{a}{x}-1\right)-\left(\frac{a}{y}-1\right)}{p-q} = \frac{\left(\frac{a}{y}-1\right)-\left(\frac{a}{z}-1\right)}{q-r} \quad \text{(by law of proportion)}$$

$$\Rightarrow \frac{\frac{a}{x}-\frac{a}{y}}{k} = \frac{\frac{a}{y}-\frac{a}{z}}{k} \quad \{\text{from (i)}\}$$

$$\Rightarrow a\left(\frac{1}{x}-\frac{1}{y}\right) = a\left(\frac{1}{y}-\frac{1}{z}\right) \Rightarrow \frac{1}{x}-\frac{1}{y} = \frac{1}{y}-\frac{1}{z}$$

$$\therefore \frac{2}{y} = \frac{1}{x} + \frac{1}{z}$$

$$\therefore \frac{1}{x}, \frac{1}{y}, \frac{1}{z} \text{ are in A.P.}$$

Hence x, y, z are in H.P.

Do yourself - 4 :

1. If the 7th term of a H.P. is 8 and the 8th term is 7. Then find the 28th term.
2. In a H.P., if 5th term is 6 and 3rd term is 10. Find the 2nd term.
3. If the pth, qth and rth terms of a H.P. are a, b, c respectively, then prove that $\frac{q-r}{a} + \frac{r-p}{b} + \frac{p-q}{c} = 0$
4. Find the fourth term in each of the following Harmonic progressions :
(a) $2, 2\frac{1}{2}, 3\frac{1}{3}, \dots$ (b) $1\frac{1}{2}, 2, 3, \dots$ (c) $3\frac{3}{5}, 6, 18, \dots$
5. If a, b, c be in H.P., show that $a : a - b = a + c : a - c$.
6. If the mth term of a H.P. be equal to n, and the nth term be equal to m, prove that the (m + n)th term is equal to $\frac{mn}{m+n}$.
7. The pth term T_p of H.P. is $q(p+q)$ and qth term T_q is $p(p+q)$ when $p > 2, q > 2 (p \neq q)$. Prove that
(a) $T_{p+q} = pq$; (b) $T_{pq} = p+q$; (c) $T_{p+q} > T_{pq}$
8. If the 10th term of a HP is 21 and 21st term of the same HP is 10, then find the 210th term.
9. If a, b, c, d, e be 5 numbers such that a, b, c are in AP ; b, c, d are in GP & c, d, e are in HP then :
(i) Prove that a, c, e are in GP.
(ii) Prove that $e = (2b - a)^2/a$
(iii) If $a = 2$ & $e = 18$, find all possible values of b, c, d.

7. MEANS

(a) ARITHMETIC MEAN :

If three terms are in A.P. then the middle term is called the A.M. between the other two, so if

a, b, c are in A.P., b is A.M. of a & c. So A.M. of a and c = $\frac{a+c}{2} = b$.

n-ARITHMETIC MEANS BETWEEN TWO NUMBERS :

If a, b be any two given numbers & $A_1, A_2, \dots, A_n$, b are in AP, then $A_1, A_2, \dots, A_n$ are the 'n' A.M.'s between a & b then. $A_1 = a + d, A_2 = a + 2d, \dots, A_n = a + nd$ or $b - d$, where

$$d = \frac{b-a}{n+1}$$

$$\Rightarrow A_1 = a + \frac{b-a}{n+1}, A_2 = a + \frac{2(b-a)}{n+1}, \dots$$

Note: Sum of n A.M's inserted between a & b is equal to n times the single A.M. between a & b

$$\text{i.e. } \sum_{r=1}^n A_r = nA \text{ where } A \text{ is the single A.M. between } a \text{ \& } b.$$

(b) GEOMETRIC MEAN :

If a, b, c are in G.P., then b is the G.M. between a & c , $b^2 = ac$. So G.M. of a and $c = \sqrt{ac} = b$

n-GEOMETRIC MEANS BETWEEN TWO NUMBERS :

If a, b are two given positive numbers & $a, G_1, G_2, \dots, G_n, b$ are in G.P. Then $G_1, G_2, G_3, \dots, G_n$ are 'n' G.Ms between a & b . where $b = ar^{n+1} \Rightarrow r = (b/a)^{1/(n+1)}$

$$\begin{aligned} G_1 &= a(b/a)^{1/(n+1)}, & G_2 &= a(b/a)^{2/(n+1)}, & \dots, & G_n &= a(b/a)^{n/(n+1)} \\ &= ar, & &= ar^2, & \dots, & &= ar^n = b/r \end{aligned}$$

Note : The product of n G.Ms between a & b is equal to n^{th} power of the single G.M. between

a & b i.e. $\prod_{r=1}^n G_r = (G)^n$ where G is the single G.M. between a & b

(c) HARMONIC MEAN :

If a, b, c are in H.P., then b is H.M. between a & c . So H.M. of a and $c = \frac{2ac}{a+c} = b$.

Insertion of 'n' HM's between a and b :

$a, H_1, H_2, H_3, \dots, H_n, b \rightarrow$ H.P

$$\frac{1}{a}, \frac{1}{H_1}, \frac{1}{H_2}, \frac{1}{H_3}, \dots, \frac{1}{H_n}, \frac{1}{b} \rightarrow \text{A.P.}$$

$$\frac{1}{b} = \frac{1}{a} + (n+1)D \Rightarrow D = \frac{\frac{1}{b} - \frac{1}{a}}{n+1}$$

$$\frac{1}{H_n} = \frac{1}{a} + n \left(\frac{\frac{1}{b} - \frac{1}{a}}{n+1} \right)$$

Important note :

- (i) If A, G, H , are respectively A.M., G.M., H.M. between two positive number a & b then
 (a) $G^2 = AH$ (A, G, H constitute a GP) (b) $A \geq G \geq H$ (c) $A = G = H \Leftrightarrow a = b$
 (ii) Let $a_1, a_2, \dots, a_n$ be n positive real numbers, then we define their arithmetic mean (A), geometric

mean (G) and harmonic mean (H) as $A = \frac{a_1 + a_2 + \dots + a_n}{n}$

$$G = (a_1 a_2 \dots a_n)^{1/n} \text{ and } H = \frac{n}{\left(\frac{1}{a_1} + \frac{1}{a_2} + \frac{1}{a_3} + \dots + \frac{1}{a_n} \right)}$$

It can be shown that $A \geq G \geq H$. Moreover equality holds at either place if and only if

$$a_1 = a_2 = \dots = a_n$$

Illustration 13 : If $2x^3 + ax^2 + bx + 4 = 0$ (a and b are positive real numbers) has 3 real roots, then prove that $a + b \geq 6(2^{1/3} + 4^{1/3})$.

Solution : Let α, β, γ be the roots of $2x^3 + ax^2 + bx + 4 = 0$. Given that all the coefficients are positive, so all the roots will be negative.

$$\text{Let } \alpha_1 = -\alpha, \alpha_2 = -\beta, \alpha_3 = -\gamma \Rightarrow \alpha_1 + \alpha_2 + \alpha_3 = \frac{a}{2}$$

$$\alpha_1\alpha_2 + \alpha_2\alpha_3 + \alpha_3\alpha_1 = \frac{b}{2}$$

$$\alpha_1\alpha_2\alpha_3 = 2$$

Applying AM $\geq$ GM, we have

$$\frac{\alpha_1 + \alpha_2 + \alpha_3}{3} \geq (\alpha_1\alpha_2\alpha_3)^{1/3} \Rightarrow a \geq 6 \times 2^{1/3}$$

$$\text{Also } \frac{\alpha_1\alpha_2 + \alpha_2\alpha_3 + \alpha_1\alpha_3}{3} \geq (\alpha_1\alpha_2\alpha_3)^{2/3} \Rightarrow b \geq 6 \times 4^{1/3}$$

Therefore $a + b \geq 6(2^{1/3} + 4^{1/3})$.

Illustration 14 : If $a_i > 0 \forall i \in N$ such that $\prod_{i=1}^n a_i = 1$,

then prove that $(1 + a_1)(1 + a_2)(1 + a_3) \dots (1 + a_n) \geq 2^n$

Solution : Using A.M. $\geq$ G.M.

$$1 + a_1 \geq 2\sqrt{a_1}$$

$$1 + a_2 \geq 2\sqrt{a_2}$$

$\vdots$

$$1 + a_n \geq 2\sqrt{a_n} \Rightarrow (1 + a_1)(1 + a_2) \dots (1 + a_n) \geq 2^n(a_1 a_2 a_3 \dots a_n)^{1/2}$$

$$\text{As } a_1 a_2 a_3 \dots a_n = 1$$

$$\text{Hence } (1 + a_1)(1 + a_2) \dots (1 + a_n) \geq 2^n.$$

Illustration 15 : If a, b, x, y are positive natural numbers such that $\frac{1}{x} + \frac{1}{y} = 1$ then prove that $\frac{a^x}{x} + \frac{b^y}{y} \geq ab$.

Solution : Consider the positive numbers $a^x, a^x, \dots, y$ times and $b^y, b^y, \dots, x$ times
For all these numbers,

$$\text{AM} = \frac{\{a^x + a^x + \dots y \text{ times}\} + \{b^y + b^y + \dots x \text{ times}\}}{x + y} = \frac{ya^x + xb^y}{(x + y)}$$

$$\text{GM} = \left\{ (a^x \cdot a^x \dots y \text{ times})(b^y \cdot b^y \dots x \text{ times}) \right\}^{\frac{1}{(x+y)}} = \left[(a^{xy}) \cdot (b^{xy}) \right]^{\frac{1}{(x+y)}} = (ab)^{\frac{xy}{(x+y)}}$$

$$\text{As } \frac{1}{x} + \frac{1}{y} = 1, \frac{x+y}{xy} = 1, \text{ i.e., } x + y = xy$$

$$\text{So using AM} \geq \text{GM } \frac{ya^x + xb^y}{x + y} \geq (ab)^{\frac{xy}{x+y}}$$

$$\therefore \frac{ya^x + xb^y}{xy} \geq ab \text{ or } \frac{a^x}{x} + \frac{b^y}{y} \geq ab.$$

Do yourself - 5 :

1. If $\frac{a^n + b^n}{a^{n-1} + b^{n-1}}$ is the G.M. between distinct positive real numbers a and b then find the value of 'n'.
2. If b is the harmonic mean between a and c , then prove that $\frac{1}{b-a} + \frac{1}{b-c} = \frac{1}{a} + \frac{1}{c}$.
3. Insert 19 arithmetic means between $\frac{1}{4}$ and $-9\frac{3}{4}$.
4. Insert 17 arithmetic means between $3\frac{1}{2}$ and $-41\frac{1}{2}$.
5. Insert 3 geometric means between $2\frac{1}{4}$ and $\frac{4}{9}$.
6. Insert 5 geometric means between $3\frac{5}{9}$ and $40\frac{1}{2}$.
7. Insert two harmonic means between 5 and 11.
8. Insert four harmonic means between $\frac{2}{3}$ and $\frac{2}{13}$.
9. If between any two quantities there be inserted two arithmetic means A_1, A_2 ; two geometric means G_1, G_2 ; and two harmonic means H_1, H_2 ; show that $G_1 G_2 : H_1 H_2 = A_1 + A_2 : H_1 + H_2$.
10. (a) The harmonic mean of two numbers is 4. The arithmetic mean A & the geometric mean G satisfy the relation $2A + G^2 = 27$. Find the two numbers.
(b) The AM of two numbers exceeds their GM by 15 & HM by 27. Find the numbers.
11. If one AM 'a' and two GM's p and q be inserted between any two given numbers then show that $p^3 + q^3 = 2apq$.
12. Given that the cubic $ax^3 - ax^2 + 9bx - b = 0$ ($a \neq 0$) has all three positive roots. Find the harmonic mean of the roots independent of a and b , hence deduce that the roots are all equal. Find also the minimum value of $(a+b)$ if a and $b \in \mathbb{N}$.
13. If $A_1, A_2, A_3, \dots, A_{51}$ are arithmetic means inserted between the numbers a and b , then find the value of $\left(\frac{b + A_{51}}{b - A_{51}} \right) - \left(\frac{A_1 + a}{A_1 - a} \right)$.

8. ARITHMETICO - GEOMETRIC SERIES :

A series, each term of which is formed by multiplying the corresponding term of an A.P. & G.P. is called the Arithmetico-Geometric Series, e.g. $1 + 3x + 5x^2 + 7x^3 + \dots$

Here 1, 3, 5, are in A.P. & 1, x , x^2 , x^3 are in G.P.

(a) SUM OF N TERMS OF AN ARITHMETICO-GEOMETRIC SERIES :

$$\text{Let } S_n = a + (a+d)r + (a+2d)r^2 + \dots + [a+(n-1)d]r^{n-1}$$

$$\text{then } S_n = \frac{a}{1-r} + \frac{dr(1-r^{n-1})}{(1-r)^2} - \frac{[a+(n-1)d]r^n}{1-r}, \quad r \neq 1$$

(b) SUM TO INFINITY :

$$\text{If } 0 < |r| < 1 \text{ \& } n \rightarrow \infty, \text{ then } \lim_{n \rightarrow \infty} r^n = 0, S_\infty = \frac{a}{1-r} + \frac{dr}{(1-r)^2}$$

Illustration 16 : Find the sum of series $4 - 9x + 16x^2 - 25x^3 + 36x^4 - 49x^5 + \dots \infty, |x| < 1$

Solution : Let $S = 4 - 9x + 16x^2 - 25x^3 + 36x^4 - 49x^5 + \dots \infty$

$$-Sx = -4x + 9x^2 - 16x^3 + 25x^4 - 36x^5 + \dots \infty$$

On subtraction, we get

$$S(1+x) = 4 - 5x + 7x^2 - 9x^3 + 11x^4 - 13x^5 + \dots \infty$$

$$-S(1+x)x = -4x + 5x^2 - 7x^3 + 9x^4 - 11x^5 + \dots \infty$$

On subtraction, we get

$$S(1+x)^2 = 4 - x + 2x^2 - 2x^3 + 2x^4 - 2x^5 + \dots \infty$$

$$= 4 - x + 2x^2(1 - x + x^2 - \dots \infty) = 4 - x + \frac{2x^2}{1+x} = \frac{4+3x+x^2}{1+x}$$

$$S = \frac{4+3x+x^2}{(1+x)^3} \quad \text{Ans.}$$

Illustration 17 : Find the sum of series upto n terms $\left(\frac{2n+1}{2n-1}\right) + 3\left(\frac{2n+1}{2n-1}\right)^2 + 5\left(\frac{2n+1}{2n-1}\right)^3 + \dots$

Solution : For $x \neq 1$, let

$$S = x + 3x^2 + 5x^3 + \dots + (2n-3)x^{n-1} + (2n-1)x^n \quad \dots (i)$$

$$\Rightarrow xS = x^2 + 3x^3 + \dots + (2n-5)x^{n-1} + (2n-3)x^n + (2n-1)x^{n+1} \quad \dots (ii)$$

Subtracting (ii) from (i), we get

$$(1-x)S = x + 2x^2 + 2x^3 + \dots + 2x^{n-1} + 2x^n - (2n-1)x^{n+1}$$

$$= x + \frac{2x^2(1-x^{n-1})}{1-x} - (2n-1)x^{n+1}$$

$$= \frac{x}{1-x} [1-x+2x-2x^n-(2n-1)x^n+(2n-1)x^{n+1}]$$

$$\Rightarrow S = \frac{x}{(1-x)^2} [(2n-1)x^{n+1} - (2n+1)x^n + 1 + x]$$

$$\text{Thus } \left(\frac{2n+1}{2n-1}\right) + 3\left(\frac{2n+1}{2n-1}\right)^2 + \dots + (2n-1)\left(\frac{2n+1}{2n-1}\right)^n$$

$$= \left(\frac{2n+1}{2n-1}\right) \left(\frac{2n-1}{2}\right)^2 \left[(2n-1)\left(\frac{2n+1}{2n-1}\right)^{n+1} - (2n+1)\left(\frac{2n+1}{2n-1}\right)^n + 1 + \frac{2n+1}{2n-1} \right]$$

$$= \frac{4n^2-1}{4} \cdot \frac{4n}{2n-1} = n(2n+1)$$

Ans.**Do yourself - 6 :**

- Find sum to n terms of the series $3 + 5 \times \frac{1}{4} + 7 \times \frac{1}{4^2} + \dots$
- sum $1 + 2a + 3a^2 + 4a^3 + \dots$ to n terms.
- sum $1 + \frac{3}{4} + \frac{7}{16} + \frac{15}{64} + \frac{31}{256} + \dots$ to infinity.
- sum $\frac{2}{3} + \frac{3}{3^2} + \frac{2}{3^3} + \frac{3}{3^4} + \frac{2}{3^5} + \frac{3}{3^6} + \dots$ to infinity.
- Find the sum of n terms of the series the r^{th} term of which is $(2r+1)2^r$.
- Find the sum of the infinite series $\frac{1.3}{2} + \frac{3.5}{2^2} + \frac{5.7}{2^3} + \frac{7.9}{2^4} + \dots \infty$.

9. SIGMA NOTATIONS (Σ)**THEOREMS :**

$$(a) \sum_{r=1}^n (a_r \pm b_r) = \sum_{r=1}^n a_r \pm \sum_{r=1}^n b_r \quad (b) \sum_{r=1}^n k a_r = k \sum_{r=1}^n a_r \quad (c) \sum_{r=1}^n k = nk ; \text{ where } k \text{ is a constant.}$$

10. RESULTS

- (a) $\sum_{r=1}^n r = \frac{n(n+1)}{2}$ (sum of the first n natural numbers)
- (b) $\sum_{r=1}^n r^2 = \frac{n(n+1)(2n+1)}{6}$ (sum of the squares of the first n natural numbers)
- (c) $\sum_{r=1}^n r^3 = \frac{n^2(n+1)^2}{4} = \left[\sum_{r=1}^n r \right]^2$ (sum of the cubes of the first n natural numbers)
- (d) $\sum_{r=1}^n r^4 = \frac{n}{30}(n+1)(2n+1)(3n^2+3n-1)$
- (e) $\sum_{r=1}^n (2r-1) = n^2$ (sum of first n odd natural numbers)
- (f) $\sum_{r=1}^n 2r = n(n+1)$ (sum of first n even natural numbers)

Note :

If n^{th} term of a sequence is given by $T_n = an^3 + bn^2 + cn + d$ where a, b, c, d are constants, then sum of n terms $S_n = \sum T_n = a\sum n^3 + b\sum n^2 + c\sum n + \sum d$

This can be evaluated using the above results.

Illustration 18 : Find the sum of the series to n terms whose n^{th} terms is $3n + 2$.

Solution : $S_n = \sum T_n = \sum (3n + 2) = 3\sum n + \sum 2 = \frac{3(n+1)n}{2} + 2n = \frac{n}{2}(3n + 7)$

Illustration 19 : If $T_k = k^3 + 3^k$, then find $\sum_{k=1}^n T_k$.

Solution : $\sum_{k=1}^n T_k = \sum_{k=1}^n k^3 + \sum_{k=1}^n 3^k = \left(\frac{n(n+1)}{2} \right)^2 + \frac{3(3^n - 1)}{3 - 1} = \left(\frac{n(n+1)}{2} \right)^2 + \frac{3}{2}(3^n - 1)$

Illustration 20 : Find the value of the expression $\sum_{i=1}^n \sum_{j=1}^i \sum_{k=1}^j 1$

Solution :
$$\begin{aligned} \sum_{i=1}^n \sum_{j=1}^i \sum_{k=1}^j 1 &= \sum_{i=1}^n \sum_{j=1}^i j = \sum_{i=1}^n \frac{i(i+1)}{2} = \frac{1}{2} \left[\sum_{i=1}^n i^2 + \sum_{i=1}^n i \right] = \frac{1}{2} \left[\frac{n(n+1)(2n+1)}{6} + \frac{n(n+1)}{2} \right] \\ &= \frac{n(n+1)}{12} [2n+1+3] = \frac{n(n+1)(n+2)}{6} \end{aligned}$$

Illustration 21 : Sum up to 16 terms of the series $\frac{1^3}{1} + \frac{1^3 + 2^3}{1+3} + \frac{1^3 + 2^3 + 3^3}{1+3+5} + \dots$ is

(A) 450

(B) 456

(C) 446

(D) none of these

Solution :

$$t_n = \frac{1^3 + 2^3 + 3^3 + \dots + n^3}{1 + 3 + 5 + \dots (2n-1)} = \frac{\left\{ \frac{n(n+1)}{2} \right\}^2}{\frac{n}{2} \{2 + 2(n-1)\}} = \frac{\frac{n^2(n+1)^2}{4}}{n^2} = \frac{(n+1)^2}{4} = \frac{n^2}{4} + \frac{n}{2} + \frac{1}{4}$$

$$\therefore S_n = \sum t_n = \frac{1}{4} \sum n^2 + \frac{1}{2} \sum n + \frac{1}{4} \sum 1 = \frac{1}{4} \cdot \frac{n(n+1)(2n+1)}{6} + \frac{1}{2} \cdot \frac{n(n+1)}{2} + \frac{1}{4} \cdot n$$

$$\therefore S_{16} = \frac{16 \cdot 17 \cdot 33}{24} + \frac{16 \cdot 17}{4} + \frac{16}{4} = 446$$

Ans. (C)

11. METHOD OF DIFFERENCE :

Some times the n^{th} term of a sequence or a series can not be determined by the method, we have discussed earlier. So we compute the difference between the successive terms of given sequence for obtained the n^{th} terms.

If $T_1, T_2, T_3, \dots, T_n$ are the terms of a sequence then some times the terms $T_2 - T_1, T_3 - T_2, \dots$ constitute an AP/GP. n^{th} term of the series is determined & the sum to n terms of the sequence can easily be obtained.

Case 1 :

(a) If difference series are in A.P., then

Let $T_n = an^2 + bn + c$, where a, b, c are constant

(b) If difference of difference series are in A.P.

Let $T_n = an^3 + bn^2 + cn + d$, where a, b, c, d are constant

Case 2 :

- (a) If difference are in G.P., then

Let $T_n = ar^n + b$, where r is common ratio & a, b are constant

- (b) If difference of difference are in G.P., then

Let $T_n = ar^n + bn + c$, where r is common ratio & a, b, c are constant

Determine constant by putting $n = 1, 2, 3, \dots, n$ and putting the value of $T_1, T_2, T_3, \dots$

and sum of series $(S_n) = \sum T_n$

Illustration 22 : Find the sum of n terms of the series $3 + 7 + 14 + 24 + 37 + \dots$

Solution : Clearly here the differences between the successive terms are

$7 - 3, 14 - 7, 24 - 14, \dots$ i.e. $4, 7, 10, 13, \dots$, which are in A.P.

Let $S = 3 + 7 + 14 + 24 + \dots + T_n$

$S = 3 + 7 + 14 + \dots + T_{n-1} + T_n$

Subtracting, we get

$$0 = 3 + [4 + 7 + 10 + 13 + \dots (n-1) \text{ terms}] - T_n$$

$\therefore T_n = 3 + S_{n-1}$ of an A.P. whose $a = 4$ and $d = 3$.

$$\therefore T_n = 3 + \left(\frac{n-1}{2} \right) (2 \cdot 4 + (n-2)3) = \frac{6 + (n-1)(3n+2)}{4} \text{ or, } T_n = \frac{1}{2} (3n^2 - n + 4)$$

Now putting $n = 1, 2, 3, \dots, n$ and adding

$$\therefore S_n = \frac{1}{2} [3 \sum n^2 - \sum n + 4n] = \frac{1}{2} \left[3 \frac{n(n+1)(2n+1)}{6} - \frac{n(n+1)}{2} + 4n \right] = \frac{n}{2} (n^2 + n + 4) \text{ Ans.}$$

Aliter Method :

Let $T_n = an^2 + bn + c$

$$\text{Now, } T_1 = 3 = a + b + c \quad \dots(i)$$

$$T_2 = 7 = 4a + 2b + c \quad \dots(ii)$$

$$T_3 = 14 = 9a + 3b + c \quad \dots(iii)$$

Solving (i), (ii) & (iii) we get

$$a = \frac{3}{2}, b = -\frac{1}{2} \text{ \& } c = 2 \quad \therefore T_n = \frac{1}{2} (3n^2 - n + 4)$$

$$\Rightarrow S_n = \sum T_n = \frac{1}{2} [3 \sum n^2 - \sum n + 4n] = \frac{1}{2} \left[3 \frac{n(n+1)(2n+1)}{6} - \frac{n(n+1)}{2} + 4n \right] = \frac{n}{2} (n^2 + n + 4)$$

Ans.

Illustration 23 : Find the sum of n-terms of the series $1 + 4 + 10 + 22 + \dots$

Solution : Let $S = 1 + 4 + 10 + 22 + \dots + T_n$ (i)

$$S = 1 + 4 + 10 + \dots + T_{n-1} + T_n \text{ (ii)}$$

$$(i) - (ii) \Rightarrow T_n = 1 + (3 + 6 + 12 + \dots + T_n - T_{n-1})$$

$$T_n = 1 + 3 \left(\frac{2^{n-1} - 1}{2 - 1} \right)$$

$$T_n = 3 \cdot 2^{n-1} - 2$$

$$\text{So } S_n = \sum T_n = 3 \sum 2^{n-1} - \sum 2$$

$$= 3 \left(\frac{2^n - 1}{2 - 1} \right) - 2n = 3 \cdot 2^n - 2n - 3$$

Ans.

Aliter Method :

Let $T_n = ar^n + b$, where $r = 2$

$$\text{Now } T_1 = 1 = ar + b \text{(i)}$$

$$T_2 = 4 = ar^2 + b \text{(ii)}$$

Solving (i) & (ii), we get

$$a = \frac{3}{2}, b = -2$$

$$\therefore T_n = 3 \cdot 2^{n-1} - 2$$

$$\Rightarrow S_n = \sum T_n = 3 \sum 2^{n-1} - \sum 2$$

$$= 3 \left(\frac{2^n - 1}{2 - 1} \right) - 2n = 3 \cdot 2^n - 2n - 3$$

Ans.

Illustration 24 : Find the general term and sum of n terms of the series

$$1 + 5 + 19 + 49 + 101 + 181 + 295 + \dots$$

Solution : The sequence of difference between successive term 4, 14, 30, 52, 80

The sequence of the second order difference is 10, 16, 22, 28, clearly it is an A.P.

So, let n^{th} term

$$T_n = an^3 + bn^2 + cn + d$$

$$a + b + c + d = 1 \quad \dots (i)$$

$$8a + 4b + 2c + d = 5 \quad \dots (ii)$$

$$27a + 9b + 3c + d = 19 \quad \dots (iii)$$

$$64a + 16b + 4c + d = 49 \quad \dots (iv)$$

from (i), (ii), (iii) & (iv)

$$a = 1, b = -1, c = 0, d = 1 \Rightarrow T_n = n^3 - n^2 + 1$$

$$\therefore S_n = \sum (n^3 - n^2 + 1) = \left(\frac{n(n+1)}{2} \right)^2 - \frac{n(n+1)(2n+1)}{6} + n = \frac{n(n^2-1)(3n+2)}{12} + n$$

Illustration 25 : If $\sum_{r=1}^n T_r = \frac{n}{8}(n+1)(n+2)(n+3)$, then find $\sum_{r=1}^n \frac{1}{T_r}$.

Solution :

$$\therefore T_n = S_n - S_{n-1}$$

$$= \sum_{r=1}^n T_r - \sum_{r=1}^{n-1} T_r = \frac{n(n+1)(n+2)(n+3)}{8} - \frac{(n-1)n(n+1)(n+2)}{8} = \frac{n(n+1)(n+2)}{8} [(n+3) - (n-1)]$$

$$T_n = \frac{n(n+1)(n+2)}{8} (4) = \frac{n(n+1)(n+2)}{2}$$

$$\Rightarrow \frac{1}{T_n} = \frac{2}{n(n+1)(n+2)} = \frac{(n+2) - n}{n(n+1)(n+2)} = \frac{1}{n(n+1)} - \frac{1}{(n+1)(n+2)} \quad \dots (i)$$

$$\text{Let } V_n = \frac{1}{n(n+1)}$$

$$\therefore \frac{1}{T_n} = V_n - V_{n+1}$$

Putting $n = 1, 2, 3, \dots, n$

$$\Rightarrow \frac{1}{T_1} + \frac{1}{T_2} + \frac{1}{T_3} + \dots + \frac{1}{T_n} = (V_1 - V_{n+1}) \Rightarrow \sum_{r=1}^n \frac{1}{T_r} = \frac{n^2 + 3n}{2(n+1)(n+2)}$$

Illustration 26 : Find the sum of n terms of the series $1 \cdot 3 \cdot 5 + 3 \cdot 5 \cdot 7 + 5 \cdot 7 \cdot 9 + \dots$

Solution : The n^{th} term is $(2n-1)(2n+1)(2n+3)$

$$T_n = (2n-1)(2n+1)(2n+3)$$

$$T_n = \frac{1}{8}(2n-1)(2n+1)(2n+3)\{(2n+5)-(2n-3)\}$$

$$= \frac{1}{8}(V_n - V_{n-1}) \text{ [Let } V_n = (2n-1)(2n+1)(2n+3)(2n+5)]$$

$$S_n = \sum T_n = \frac{1}{8}[V_n - V_0]$$

$$\therefore S_n = \frac{(2n-1)(2n+1)(2n+3)(2n+5)}{8} + \frac{15}{8} = n(2n^3 + 8n^2 + 7n - 2) \quad \text{Ans.}$$

Illustration 27 : Find the natural number 'a' for which $\sum_{k=1}^n f(a+k) = 16(2^n - 1)$, where the function f satisfied

$f(x+y) = f(x) \cdot f(y)$ for all natural number x, y and further $f(1) = 2$.

Solution : It is given that

$$f(x+y) = f(x) \cdot f(y) \text{ and } f(1) = 2$$

$$f(1+1) = f(1) \cdot f(1) \Rightarrow f(2) = 2^2, f(1+2) = f(1) \cdot f(2) \Rightarrow f(3) = 2^3, f(2+2) = f(2)$$

$$f(2) \Rightarrow f(4) = 2^4$$

$$\text{Similarly } f(k) = 2^k \text{ and } f(a) = 2^a$$

$$\text{Hence, } \sum_{k=1}^n f(a+k) = \sum_{k=1}^n f(a)f(k) = f(a) \sum_{k=1}^n f(k) = 2^a \sum_{k=1}^n 2^k = 2^a \{2^1 + 2^2 + \dots + 2^n\}$$

$$= 2^a \left\{ \frac{2(2^n - 1)}{2 - 1} \right\} = 2^{a+1}(2^n - 1)$$

$$\text{But } \sum_{k=1}^n f(a+k) = 16(2^n - 1)$$

$$2^{a+1}(2^n - 1) = 16(2^n - 1)$$

$$\therefore 2^{a+1} = 2^4$$

$$\therefore a+1 = 4 \Rightarrow a = 3$$

Ans.

Do yourself - 7 :

1. Find the sum of the series upto n terms $1 + \frac{1+2}{2} + \frac{1+2+3}{3} + \frac{1+2+3+4}{4} + \dots$
2. Find the sum of 'n' terms of the series whose n^{th} term is
 - (a) $3n^2 + 2n$
 - (b) $3n^2 - n$
 - (c) $n^3 + \frac{3}{2}n$
 - (d) $n(n+2)$
 - (e) $n^2(2n+3)$
 - (f) $3^n - 2^n$
 - (g) $3(4^n + 2n^2) - 4n^3$
3. Sum the following series to n terms :
 - (a) $1 \cdot 4 \cdot 7 + 2 \cdot 5 \cdot 8 + 3 \cdot 6 \cdot 9 + \dots$
 - (b) $1 \cdot 5 \cdot 9 + 2 \cdot 6 \cdot 10 + 3 \cdot 7 \cdot 11 + \dots$
4. Find the n^{th} term and the sum of n terms of the series :
 - (a) 4, 14, 30, 52, 80, 114,
 - (b) 9, 16, 29, 54, 103,
5. Find the n^{th} term and the sum to n terms of the sequence :
 - (i) $1 + 5 + 13 + 29 + 61 + \dots$
 - (ii) $6 + 13 + 22 + 33 + \dots$
6. Sum the following series to n terms and to infinity (where it is finite and defined) :
 - (i) $\frac{1}{1 \cdot 4 \cdot 7} + \frac{1}{4 \cdot 7 \cdot 10} + \frac{1}{7 \cdot 10 \cdot 13} + \dots$
 - (ii) $\sum_{r=1}^n r(r+1)(r+2)(r+3)$
 - (iii) $\sum_{r=1}^n \frac{1}{4r^2 - 1}$
7. The value of $1^2 + 3^2 + 5^2 + \dots + 25^2$ is :

(A) 1728	(B) 1456	(C) 2925	(D) 1469
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8. The sum of the series : $(2)^2 + 2(4)^2 + 3(6)^2 + \dots$ upto 10 terms is :

(A) 11300	(B) 12100	(C) 12300	(D) 11200
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9. The value of $(1 \cdot 1^2 + 3 \cdot 2^2 + 5 \cdot 3^2 + \dots + \text{upto 10 terms})$ is equal to :

(A) 6050	(B) 5965	(C) 5665	(D) 5385
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